

Critical Thinking Questions - Chapter 4

1. Write appropriate statement for each of the following

- a)

```
if (grade >= 90) {  
    System.out.println("Great Job!");  
}
```
- b)

```
if (number < 20 || number > 50) {  
    System.out.println("Error");  
}
```
- c)

```
if (y < 100) {  
    Y = y + 2;  
}
```

2. Write an else-if statement that displays one of the three messages:

```
if (num1 > num2) {  
    System.out.println("First number is larger.");  
}  
else if (num2 > num1) {  
    System.out.println("Second number is larger.");  
}  
else {  
    System.out.println("numbers are equal");  
}
```

3. Which is the appropriate word, odd or even for the blanks below?

- a) Even, Odd (respectively)
- b)

```
Switch (num % 2) {  
    case 0:  
        System.out.println("even number");  
        break;  
    case 1:  
        System.out.println("odd number");  
        break;  
}
```

4. Write statements that use Math.random() to generate numbers for different situations:

- a) `int num = (int)(Math.random() * 50) + 1;`
- b) `int num = (int)(Math.random() * 81) 20;`
- c) `double num = Math.random() * (20 - 10) + 10;`

5. Identify any logic problems in the statement below:

Ages 18 and 65 are missing. When writing ranges ensure no values are skipped.

6. Determine if each of the following expressions evaluates to true or false:

- a) True
- b) False
- c) True
- d) True
- e) True
- f) True
- g) True

8. Determine if each of the following are true or false, if false explain:

- a) True
- b) False
- c) False
- d) True
- e) False
- f) True
- g) True
- h) False
- i) True
- j) False